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The Editorial Office
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Dear Editors,

We submit our manuscript “**UzABSA-LLM: Parameter-Efficient Fine-Tuning and Multi-Domain Annotation for Uzbek Aspect-Based Sentiment Analysis**” for consideration as an original research article in *Big Data and Cognitive Computing*, for the Special Issue “Natural Language Processing Applications in Big Data”.

Significance. The paper does two things: it establishes what parameter-efficient fine-tuning delivers for Uzbek aspect-based sentiment analysis (ABSA), and it turns the resulting model into a pipeline that converts 5,038 business reviews across 23 domains into a quality-scored annotated resource. Uzbek, a Turkic language with over 38 million speakers, is under-served by multilingual models owing to sparse pre-training representation, agglutinative morphology, and script duality. This is the first controlled comparison of QLoRA-adapted open-weight 7–8B models on the UzABSA benchmark, and the first to benchmark them against monolingual Uzbek encoders.

Findings in context. Generative ABSA has concentrated on high-resource languages, while prior work on Uzbek—widely spoken across Central Asia but reasonably low-resource in terms of annotated corpora—is rule-based, classical, or encoder-based. Against zero-shot controls on the same split, QLoRA fine-tuning at least doubles exact extraction F1 and triples end-to-end pair F1. Qwen 2.5-7B reaches the best extraction F1 (0.708) at a 100% JSON parse rate, and Llama 3.1-8B leads pair F1 (0.651) and sentiment accuracy (0.925); a paired bootstrap finds no significant difference between them, which we report as such rather than declaring a winner. More consequentially for practitioners, fine-tuned 110M-parameter Uzbek encoders match the 7–8B models at aspect extraction while losing clearly on sentiment and joint prediction—a cost-capability trade-off not previously quantified for this language. All code, data, and models are publicly released.

Fit to BDCC. For user-generated text in a low-resource language, the binding constraint on large-scale analysis is not storage or compute but *annotation*: a heterogeneous review stream cannot become an analytical resource without labels that do not exist and cannot be produced by hand at scale. Our pipeline performs that conversion, and we report its economics from measured runs rather than asserting scalability: roughly 840 reviews per hour on a single GPU, with automated quality scoring of the whole corpus costing under one US dollar. The remaining cost is human, which is why the automatic layers run exhaustively while scarce native-speaker effort goes to calibrating them.

That calibration is where the work meets the journal’s cognitive-computing side. We substitute machine judgement for human judgement and treat the substitution as a claim to be tested: an LLM judge is calibrated against native-speaker ratings, and we report where it succeeds ($\rho = 0.81$ on sentiment, $\rho = 0.66$ on the tier-defining overall score) and where it does not (completeness, where it is systematically harsher than humans). The study also yields a finding about the task itself: annotators agreed on the polarity of every jointly identified aspect, whereas *what counts as the aspect span* was the dominant disagreement even between trained native speakers. Human and machine annotators diverge in the same place, and this calibration is what licenses the released dataset’s quality tiers.

Prior MDPI submission. As required, we disclose that an earlier version of this manuscript was submitted to *Computers* (MDPI) under manuscript ID **computers-4419237** and was not

accepted; this ID is also supplied in the submission system. The present version differs substantially: all models were retrained after we identified and corrected two defects in the earlier pipeline (a system-prompt serialization bug and text-level overlap between the training and evaluation splits), and we added zero-shot controls, encoder baselines, paired-bootstrap significance testing, and a native-speaker validation study. The results are new and the claims narrowed accordingly. The author list is unchanged from that submission.

We confirm that neither the manuscript nor any parts of its content are currently under consideration for publication with or published in another journal. All authors have approved the manuscript and agree with its submission to BDCC.

Thank you for considering our manuscript.

Sincerely,



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